Adaptive Extended Disturbance Observer-Based Control for Quadrotor Landing on Fast Platform

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***Abstract*—In order to land quadrotor uncrewed aerial vehicles precisely and reliably on fast platforms, complex aerodynamic disturbances can be estimated by disturbance observers with proper bandwidth and then compensated by feedforward. However, the observer bandwidth is difficult to design since both aerodynamic disturbances and measurement noises vary significantly in the landing process. In this article, an adaptive extended disturbance observer-based control scheme is proposed to improve disturbance estimation accuracy and suppress aerodynamic disturbances in the landing of quadrotors. First, the time-varying properties of aerodynamic disturbances and measurement noises are analyzed in the landing process. Then, two adaptive extended disturbance observers (AEDOs) are designed for translational and rotational subsystems, respectively with their bandwidths adaptively tuned according to the power ratio of disturbances to measurement noises. Theoretical analysis indicates AEDO has higher accuracy in disturbance estimation as well as robustness to measurement noises compared with other EDOs. Finally, block backstepping controllers are designed for the subsystems with the outputs of the AEDOs as feedforward compensation. Landing experiments on a platform of 8.3 m/s demonstrate the effectiveness of the proposed scheme in tuning the observer bandwidth adaptively according to disturbance and noise information and improving the precision of landing points.**

***Index Terms*—Adaptive observer, autonomous landing, disturbance observer, quadrotor, uncrewed aerial vehicles (UAVs).**

I. Introduction

In the landing of quadrotor uncrewed aerial vehicles (UAVs) on moving platforms, aerodynamic disturbances are dominant factors affecting the precision of landing points and thus the landing reliability [1], [2], [3]. Especially with the increase of the platform speed, both the strength and frequency content of aerodynamic disturbances will become higher due to the effect of airflow between the quadrotor and platform. These characteristics will make it difficult to suppress aerodynamic disturbances in the landing control. They will deteriorate the precision of landing points and even lead to landing failure on fast platforms. Therefore, suppressing complex aerodynamic disturbances has become one of the key issues in reliable landing control of quadrotors on fast platforms.

TABLE I
Comparison of Landing Schemes

Schemes	Ref.	Conditions	Platform Speed (m/s)	Landing Point Error (cm)
PID	[4]	Indoor	∼1.1	<10*
Backstepping control	[5]	Indoor	0.5	3.5
Adaptive control	[6]	Outdoor	≤1*	∼30
Sliding mode control	[7]	Indoor	≤1*	∼15*
Intelligent control	[8]	Indoor	∼0.3*	∼20*
HLS + robust control	[9]	Outdoor	∼5.6	∼75.1
HLS + PID	[10]	Outdoor	∼13.9	∼70*
DO + robust control	[16]	Indoor	∼0.25*	∼3*
DO + robust control	[17]	Indoor	∼0.5*	∼15*
DO + adaptive control	[18]	Outdoor	0.4*	∼4*
DO + adaptive control	[19]	Indoor	∼1*	∼5*
ESO + ADRC	[20]	Outdoor	<1*	<5*
ESO + ADRC	[21]	Indoor	1.6	∼1.5*
EDO + BC	[22]	Outdoor	∼5.6	8.5
NREDO + BC	[23]	Outdoor	∼7	7.1

Note: *Stands for the data obtained from the experimental curves in the references.

Feedback control is the most convenient solution to landing quadrotors on moving platforms, as shown in Table I [4], [5], [6], [7], [8]. However, the conventional feedback control methods can not deal with strong aerodynamic disturbances in the landing process, including PID control [4], backstepping control (BC) [5], adaptive control (AC) [6], and sliding mode control (SMC) [7]. Especially when a quadrotor approaches a fast platform, the aerodynamic disturbances will be further strengthened due to the vortices caused by the rotating propellers near the platform surface. Even with intelligent control [8], there is no significant improvement in the anti-disturbance performance.

A simple solution to this problem is to stop propellers when the quadrotor reaches a specified altitude directly above the platform, namely the hard landing scheme (HLS). As the aerodynamic disturbances can be significantly reduced by propeller

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shutdown, the quadrotor can land on the fast platform with this strategy. However, the HLS will cause uncontrollable landing points and damage to onboard equipments [9], [10].

To suppress aerodynamic disturbances in the entire landing process, they can be estimated by disturbance observers first and then compensated by feedforward. In this strategy, the key to disturbance suppression is accurate estimation [11], [12], [13], [14], [15]. Although both disturbance observer (DO) [16], [17], [18], [19] and extended state observer (ESO) [20], [21] have been employed to estimate aerodynamic disturbances in the landing on moving platforms, they are designed based on the assumption of slowly-varying disturbances. Therefore, it is difficult for them to accurately estimate complex aerodynamic disturbances in the landing on fast platforms.

To accurately estimate complex aerodynamic disturbances, extended disturbance observer (EDO) and noise reduction extended disturbance observer (NREDO) can be utilized [22], [23]. Although NREDO has better noise suppression performance compared with EDO, both of the observers typically use fixed bandwidth to tune gains. To ensure accurate estimation, the bandwidth design should consider the effects of aerodynamic disturbances and measurement noises in a balanced manner. However, aerodynamic disturbances vary with the relative height between the propellers and platform when the quadrotor is close to the platform. In fact, the measurement noises from the measurement of the relative distance between the quadrotor and platform varies with the relative distance due to the resolution limitations of the vision camera. Therefore, it is difficult to ensure accurate estimation with fixed bandwidth in the entire process of quadrotors landing on fast platforms.

For disturbance estimation in the presence of time-varying disturbances and measurement noises, many studies have been reported in other fields. For example, Kalman filter [24] and α–β–γ filter [25] have been used in object tracking to cope with time-varying disturbances and measurement noises, and their bandwidths are tuned automatically according to the statistical characteristics of disturbances and the noises. It should be noted that their estimation accuracy relies on the Gaussian distribution assumption of disturbances and noises. However, aerodynamic disturbances do not satisfy the assumption.

In summary, it remains a critical issue to suppress complex aerodynamic disturbances for quadrotor UAVs landing on fast platforms. The motivation of this article is to handle the variation of disturbances and measurement noises so as to improve estimation accuracy and suppress disturbances in the landing scenario. Inspired by the idea of adaptive bandwidth, an adaptive EDO (AEDO) based control scheme is proposed in the landing of quadrotors on fast platforms.

The main contributions of this article are listed as follows.

1) According to the characteristics of aerodynamic disturbances and measurement noises in the landing process, two AEDOs are designed for the translational and rotational subsystems of a quadrotor to estimate force and torque disturbances. With the bandwidth tuned automatically based on the power ratio of disturbances to measurement noises, AEDO can improve disturbance estimation accuracy.
2) Based on the outputs of the AEDOs as feedforward compensation, backstepping controllers are designed for the two subsystems. With the AEDO-based control scheme, precise and reliable landing can be achieved.

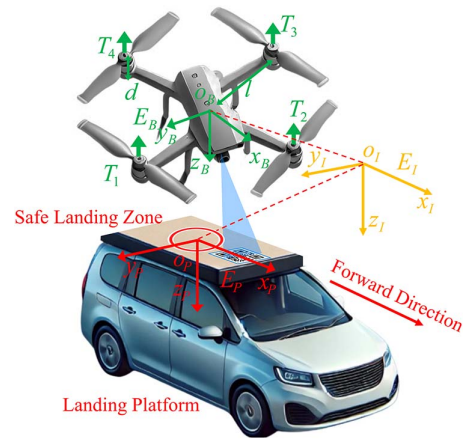

Fig. 1. Coordinate frames in quadrotor UAVs landing on moving platforms.

II. Mathematical Model and Problem Formulation

To land a quadrotor on a fast platform, it should approach the platform and land precisely on the safe landing zone. To describe the landing motion, the coordinate frames in Fig. 1 are defined: the inertial frame E_I, the platform frame E_P, and the body-fixed frame E_B. Additionally, the variables of the quadrotor are used in Fig. 1: l is the length of the arm, d denotes the distance between the x_B–y_B plane and propeller disk, and $\mathbf{l}_i = [l\cos[(3-2i)\pi/4] \;\; l\sin[(3-2i)\pi/4] - d]^T$ presents the position of the ith propeller in E_B, where $i = 1, 2, 3, 4$.

A. Description of Quadrotor Landing on Fast Platform

Define the position and velocity of the platform as $\boldsymbol{p}$ and $\boldsymbol{v}_p$, then the platform motion can be described as follows:

$$\begin{cases} \dot{\boldsymbol{p}} = \boldsymbol{v}_p \\ M\dot{\boldsymbol{v}}_p = \boldsymbol{F}_{pc} + \boldsymbol{F}_{pd} + M\boldsymbol{g}. \end{cases} \tag{1}$$

Define the position and velocity of the quadrotor as $\boldsymbol{r}$ and $\boldsymbol{v}$, and its attitude quaternion and angular velocity as $\boldsymbol{q}$ and $\boldsymbol{\omega}$, then its dynamics can be expressed as follows:

$$\begin{cases} \dot{\boldsymbol{r}} = \boldsymbol{v} \\ m\dot{\boldsymbol{v}} = \boldsymbol{F}_c + \boldsymbol{F}_d \\ \dot{\boldsymbol{q}} = \boldsymbol{q} \otimes [0 \quad \boldsymbol{\omega}]^{\mathrm{T}}/2 \\ \boldsymbol{J}\dot{\boldsymbol{\omega}} + \boldsymbol{\omega}^{\times}\boldsymbol{J}\boldsymbol{\omega} = \boldsymbol{\tau}_c + \boldsymbol{\tau}_d \end{cases} \tag{2}$$

where M and m are the masses of the platform and quadrotor. $\boldsymbol{F}_{pc}$, $\boldsymbol{F}_c$, and $\boldsymbol{\tau}_c$ are the control inputs. $\boldsymbol{F}_{pd}$, $\boldsymbol{F}_d$, and $\boldsymbol{\tau}_d$ are the lumped disturbance. $\boldsymbol{F}_c = \Sigma_{i=1}^4 \boldsymbol{R}_B^I \boldsymbol{T}_i + m\boldsymbol{g}$ and $\boldsymbol{\tau}_c = \Sigma_{i=1}^4 (\boldsymbol{\tau}_i + \boldsymbol{l}_i^{\times}\boldsymbol{T}_i)$, where $\boldsymbol{R}_B^I$ is the rotation matrix from

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E_B to E_I. $\boldsymbol{l}_i^{\times}$ is the skew-symmetric matrix for $\boldsymbol{l}_i$. Furthermore, $\boldsymbol{T}_i$ and $\boldsymbol{\tau}_i$ are the thrust and anti-torque of the ith propeller.

For (1) and (2), $\boldsymbol{F}_c$ and $\boldsymbol{\tau}_c$ need to be designed to make $\boldsymbol{r}$ converge to $\boldsymbol{p}$. However, $\boldsymbol{F}_d$ and $\boldsymbol{\tau}_d$ reduce the convergence accuracy and may even lead to unreliable landing. $\boldsymbol{F}_d$ and $\boldsymbol{\tau}_d$ contain both aerodynamic disturbances and model uncertainties, where aerodynamic disturbances are particularly complex.

B. Aerodynamic Disturbances

As aerodynamic disturbances are complex and affect the landing performance, their characteristics should be analyzed.

1) Mechanism: Aerodynamic disturbances refer to aerodynamic forces that are typically not modeled in the flight system, including translational thrust, propeller disk drag, and parasitic drag [26]. Define the velocity of the airflow as $\boldsymbol{v}_a$, and the velocity of a quadrotor relative to the airflow is expressed as $\boldsymbol{v}_r = \boldsymbol{v} + \boldsymbol{v}_a$. For the ith propeller, its translational thrust and propeller disk drag are represented as [27]

$$\boldsymbol{T}_{it} = [0 \quad 0 \quad -k_z \boldsymbol{\Omega}_i v_{rz} - k_{xy}(v_{rx}^2 + v_{ry}^2)]^{\boldsymbol{T}} \tag{3}$$

$$\boldsymbol{F}_{ib} = c_b ||\boldsymbol{T}_i + \boldsymbol{T}_{it}|| [v_{rx} \quad v_{ry} \quad 0]^T \tag{4}$$

where $\|\cdot\|$ denotes the two-norm of a vector, c_b, k_{xy}, and k_z are positive constant coefficients.

For parasitic drag and drag torque, they are expressed as [27] follows:

$$\boldsymbol{F}_p = \boldsymbol{K}_{Fp}[v_{rx}^2 \quad v_{ry}^2 \quad v_{rz}^2]^T \tag{5}$$

$$\boldsymbol{\tau}_p = \boldsymbol{K}_{\tau p}[v_{rx}^2 \quad v_{ry}^2 \quad v_{rz}^2]^T \tag{6}$$

where $\boldsymbol{K}_{Fp}$ and $\boldsymbol{K}_{\tau p}$ are drag and torque coefficient matrices.

Based on (3) to (6), the total aerodynamic force and torque disturbance $\boldsymbol{F}_a$ and $\boldsymbol{\tau}_a$ can be written as follows:

$$\boldsymbol{F}_a = \sum_{i=1}^{4} (\boldsymbol{F}_{ib} + \boldsymbol{T}_{it}) + \boldsymbol{F}_p \tag{7}$$

$$\boldsymbol{\tau}_a = \sum_{i=1}^{4} [\boldsymbol{l}_i^{\times}(\boldsymbol{F}_{ib} + \boldsymbol{T}_{it})] + \boldsymbol{\tau}_p. \tag{8}$$

2) Influencing Factors: Due to fast platform speed, wind gusts, and uneven ground effects, the disturbances in (3)–(8) become more complex.

For platform speed $\boldsymbol{v}_p$, high $\boldsymbol{v}_p$ can lead to high $\boldsymbol{v}_r$ which amplify the intensity of aerodynamic disturbances in (3)–(8).

For wind gusts, atmospheric turbulence and wind shear in them can cause transient changes in $\boldsymbol{v}_r$ [28]. When $\boldsymbol{v}_p$ increases, the quadrotor encounters the turbulence and wind shear more frequently, and this will enhance the dynamics of the $\boldsymbol{v}_r$ and that of aerodynamic disturbances in (3)–(8).

For ground effects, they are uneven in the landing scenario due to attitude tilting and incomplete projection of propellers on the platform [29]. Define the height of the ith propeller relative to the platform as z_i, and the force and torque disturbances by ground effects are denoted as [30] follows:

$$\boldsymbol{T}_{i\mathrm{GE}} = [0 \quad 0 \quad (b_i + k_i/z_i^2)||\boldsymbol{T}_i||]^{\mathrm{T}} \tag{9}$$

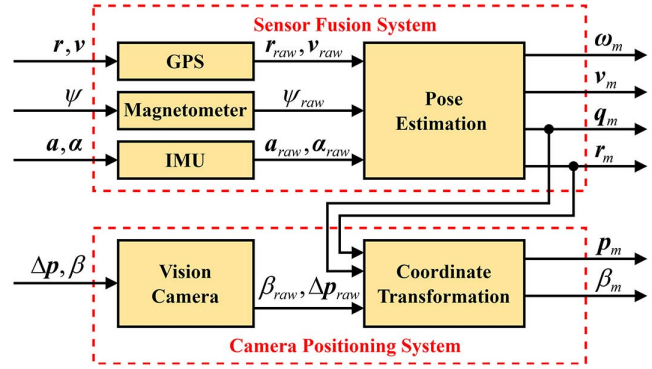

Fig. 2. Measurement procedures for motion information in the landing.

$$\boldsymbol{\tau}_{\mathrm{UGE}} = \sum_{i=1}^{4} \boldsymbol{l}_i \times \boldsymbol{T}_{i\mathrm{GE}} \tag{10}$$

where b_i and k_i are coefficients related to the attitude tilting and the incomplete projection. Under uneven ground effects, the total aerodynamic disturbance can be further expressed as $\boldsymbol{F}_a^* = \boldsymbol{F}_a + \sum_{i=1}^{4} \boldsymbol{T}_{i\mathrm{GE}}$ and $\boldsymbol{\tau}_a^* = \boldsymbol{\tau}_a + \boldsymbol{\tau}_{\mathrm{UGE}}$. When the quadrotor approaches the platform, z_i decreases, and then $\boldsymbol{T}_{i\mathrm{GE}}$ and $\boldsymbol{\tau}_{\mathrm{UGE}}$ intensify, and thus $\boldsymbol{F}_a^*$ and $\boldsymbol{\tau}_a^*$ become stronger.

Therefore, aerodynamic disturbances not only become stronger and have faster dynamics due to increased platform speed and wind gusts, but also become further strengthened with the quadrotor descent and uneven ground effects.

3) Assumption of Disturbances: Although aerodynamic disturbances in the landing scenario are complex, they come from continuous physical processes like other disturbances. Therefore, they can still be considered as the continuous and bounded lumped disturbance along with other disturbances.

Assumption 1: $\boldsymbol{F}_d$ and $\boldsymbol{\tau}_d$ are continuous and with bounded derivatives such that $||\boldsymbol{F}_d^{(j)}|| \le \gamma_F$ and $||\boldsymbol{\tau}_d^{(j)}|| \le \gamma_T$, $j = 1, 2, \ldots, n$, γ_F, and γ_T are the upper bounds.

C. Measurement Noises

To land on a fast platform, a quadrotor requires not only a sensor fusion system to measure its own pose, but also a camera positioning system to measure the platform motion [31]. As the measurement noises in both systems affect the landing performance, their characteristics should be analyzed.

Define the heading of the platform and quadrotor as β and ψ, and denote the acceleration and angular acceleration of the quadrotor as $\boldsymbol{a}$ and $\boldsymbol{\alpha}$, then the measurement procedures of both systems are shown in Fig. 2, where the subscripts "*raw*" and "*m*" represent the raw sensor data and measured value.

For the sensor fusion system, it estimates the quadrotor pose with the data from GPS, IMU, and a magnetometer [32]. Although much noises have been filtered out by the pose estimation, residual noises remains in $\boldsymbol{r}_m$, $\boldsymbol{v}_m$, $\boldsymbol{q}_m$, and $\boldsymbol{\omega}_m$.

In the camera positioning system, β and the relative distance between the quadrotor and platform $\Delta\boldsymbol{p} = \boldsymbol{r} - \boldsymbol{p}$ are measured by a vision camera according to the specific marker on the platform [33]. Due to the limited resolution of the vision camera, the increase in $\Delta\boldsymbol{p}$ will reduce the number of pixels occupied

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by the specific marker, and thus enhance the measurement noise in $\Delta \boldsymbol{p}_{\text{raw}}$. As $\boldsymbol{p}_m$ is calculated using $\boldsymbol{r}_m$, $\boldsymbol{q}_m$, β_{raw}, and $\Delta \boldsymbol{p}_{\text{raw}}$, the increase in $\Delta \boldsymbol{p}$ will increase the measurement noise in $\boldsymbol{p}_m$.

In summary, the measurement noises in the quadrotor are complex as the intensity of the measurement noises in β_m and $\boldsymbol{p}_m$ varies with $\Delta \boldsymbol{p}$. The greater the relative distance $\Delta \boldsymbol{p}$, the stronger the measurement noises in β_m and $\boldsymbol{p}_m$.

Define the measurement noises in β_m, $\boldsymbol{p}_m$, $\boldsymbol{r}_m$, $\boldsymbol{v}_m$, $\boldsymbol{q}_m$, and $\boldsymbol{\omega}_m$ as n_β, $\boldsymbol{n}_p$, $\boldsymbol{n}_r$, $\boldsymbol{n}_v$, $\boldsymbol{n}_q$, and $\boldsymbol{n}_\omega$, and these measurement can be further expressed as $\beta_m = \beta + n_\beta$, $\boldsymbol{p}_m = \boldsymbol{p} + \boldsymbol{n}_p$, $\boldsymbol{r}_m = \boldsymbol{r} + \boldsymbol{n}_r$, $\boldsymbol{v}_m = \boldsymbol{v} + \boldsymbol{n}_v$, $\boldsymbol{q}_m = \boldsymbol{n}_q^* \otimes \boldsymbol{q}$, and $\boldsymbol{\omega}_m = \boldsymbol{\omega} + \boldsymbol{n}_\omega$.

Assumption 2: n_β, $\boldsymbol{n}_p$, $\boldsymbol{n}_r$, $\boldsymbol{n}_v$, $\boldsymbol{n}_q$, and $\boldsymbol{n}_\omega$ are bounded and with bounded derivates such that $||n_\beta^{(j)}|| \le \delta_\beta$, $||\boldsymbol{n}_p^{(j)}|| \le \delta_p$, $||\boldsymbol{n}_r^{(j)}|| \le \delta_r$, $||\boldsymbol{n}_v^{(j)}|| \le \delta_v$, $||\boldsymbol{n}_q^{(j)}|| \le \delta_q$, $||\boldsymbol{n}_\omega^{(j)}|| \le \delta_\omega$, $j = 0, 1, \ldots, n$, where δ_β, δ_p, δ_r, δ_v, δ_q, and δ_ω are the upper bounds.

D. Problem Statement

The aforementioned analysis indicates that the intensity of aerodynamic disturbances increases and that of measurement noises decreases when the quadrotor descends. Moreover, the dynamics of aerodynamic disturbances become faster in the landing on fast platforms. In this case, it is difficult for disturbance observers to ensure accurate estimation with fixed bandwidth in the entire landing process. As a result, complex aerodynamic disturbances will reduce the precision of landing points and make the landing unreliable. Inspired by the characteristics of aerodynamic disturbances and measurement noises in the landing scenario, it is necessary to design disturbance observers with variable bandwidths to adaptively handle the changes in disturbances and measurement noises and design controllers with the outputs of the observers as feedforward compensation to suppress complex aerodynamic disturbances in the landing scenario.

III. AEDOs for Aerodynamic Disturbances

As the anti-disturbance and dynamic performance of feedback control are limited in the landing scenario, it is necessary to estimate and compensate for complex aerodynamic disturbances with disturbance observers. In this section, AEDOs will be designed for the translational and rotational subsystems, respectively to estimate force and torque disturbances, and their performance will be analyzed.

A. Translational AEDO

Define the states of $\boldsymbol{F}_d$ as $\boldsymbol{x}_{Fi} = \boldsymbol{F}_d^{(i-1)}$ and its state matrix as $\boldsymbol{x}_F = [\boldsymbol{x}_{F1} \quad \boldsymbol{x}_{F2} \quad \cdots \quad \boldsymbol{x}_{Fn}]^{\mathrm{T}}$, and then the nth-order model of $\boldsymbol{F}_d$ is constructed as follows:

$$\dot{\boldsymbol{x}}_F = \boldsymbol{A}_F \boldsymbol{x}_F + \boldsymbol{B}_F \boldsymbol{F}_d^{(n)} \tag{11}$$

where $\boldsymbol{A}_F = \begin{bmatrix} \boldsymbol{O}_{3(n-1)\times 3} & \boldsymbol{I}_{3(n-1)} \\ \boldsymbol{O}_{3\times 3} & \boldsymbol{O}_{3\times 3(n-1)} \end{bmatrix}$, $\boldsymbol{B}_F = \begin{bmatrix} \boldsymbol{O}_{3(n-1)\times 3} \\ \boldsymbol{I}_3 \end{bmatrix}$, $i = 1, 2, \ldots, n$.

According to (2) and $\boldsymbol{v}_m = \boldsymbol{v} + \boldsymbol{n}_v$, the virtual measurement of $\boldsymbol{F}_d$ is obtained by $\boldsymbol{F}_{dm} = m\dot{\boldsymbol{v}}_m - \boldsymbol{F}_c = \boldsymbol{F}_d + m\dot{\boldsymbol{n}}_v$. Define $\boldsymbol{n}_F = m\dot{\boldsymbol{n}}_v$ and $\boldsymbol{C}_F = [\boldsymbol{I}_3 \quad \boldsymbol{O}_{3\times 3(n-1)}]$, and then the output of (11) is written as follows:

$$\boldsymbol{y}_F = \boldsymbol{C}_F \boldsymbol{x}_F + \boldsymbol{n}_F. \tag{12}$$

For AEDO, define the estimate of $\boldsymbol{F}_d$ as $\hat{\boldsymbol{x}}_{AFi} = \hat{\boldsymbol{F}}_{dA}^{(i-1)}$, and estimate matrix as $\hat{\boldsymbol{x}}_{AF} = [\hat{\boldsymbol{F}}_{dA} \quad \hat{\boldsymbol{F}}_{dA}^{(1)} \quad \cdots \quad \hat{\boldsymbol{F}}_{dA}^{(n-1)}]^T$, and then the nth-order translational AEDO is designed as follows:

$$\begin{cases} \dot{\hat{\boldsymbol{x}}}_{AF} = \boldsymbol{A}_F \hat{\boldsymbol{x}}_{AF} + \boldsymbol{L}_{AF} \boldsymbol{C}_F (\boldsymbol{x}_F - \hat{\boldsymbol{x}}_{AF}) + \boldsymbol{L}_{AF} \boldsymbol{n}_F \\ \hat{\boldsymbol{y}}_{AF} = \boldsymbol{C}_F \hat{\boldsymbol{x}}_{AF} \end{cases} \tag{13}$$

where $\boldsymbol{L}_{AF} = [l_{AF1}\omega_{AF}\boldsymbol{I}_3 \quad l_{AF2}\omega_{AF}^2\boldsymbol{I}_3 \quad \cdots \quad l_{AFn}\omega_{AF}^n\boldsymbol{I}_3]^{\mathrm{T}}$ is the gain matrix, $l_{AFi} > 0$, ω_{AF} is the variable bandwidth, and $\hat{\boldsymbol{y}}_{AF}$ denotes the estimate of $\boldsymbol{y}_F$.

As the variations of disturbances and measurement noises are the key to the estimation accuracy in the landing process, it is necessary to adjust the variable bandwidth ω_{AF} based on disturbances and measurement noises. To establish the relationship between ω_{AF}, disturbances, and measurement noises, a unified standard should first be used to describe disturbances and measurement noises. As disturbances and measurement noises may fluctuate drastically over short periods, their power smoothed by mean filter can be used to stably characterize their properties. Define the power of $\boldsymbol{F}_d^{(n)}$ and $\boldsymbol{n}_F$ as $||\boldsymbol{F}_d^{(n)}||^2$ and $||\boldsymbol{n}_F||^2$, and the power smoothed by mean filter as P_F and P_{NF}, then P_F and P_{NF} are denoted as follows:

$$P_F = \int_{t-\Delta t}^{t} ||\boldsymbol{F}_d^{(n)}(t)||^2 dt/\Delta t \tag{14}$$

$$P_{NF} = \int_{t-\Delta t}^{t} ||\boldsymbol{n}_F(t)||^2 dt/\Delta t \tag{15}$$

where Δt is a sliding window of mean filter.

Define the power ratio between $\boldsymbol{F}_d^{(n)}$ and $\boldsymbol{n}_F$ as $\eta_F = P_F/P_{NF}$, and the adaptive tuning rule of ω_{AF} is designed as follows:

$$\omega_{AF} = \begin{cases} (K_F\eta_F)^{1/2n}, \ (K_F\eta_F)^{1/2n} > \omega_{AFm} \\ \omega_{AFm}, \text{other} \end{cases} \tag{16}$$

where $K_F = \begin{cases} 1, \ \int_{t-\Delta t}^{t} \omega_{AF}^2 dt/\Delta t < 1 \\ \int_{t-\Delta t}^{t} \omega_{AF}^2 dt/\Delta t, \text{other} \end{cases}$ is a coefficient used to accelerate the tuning speed, and $\omega_{AFm} = \min(\omega_{AF})$.

To analyze the observer stability, define the estimation error of the translational AEDO as $\tilde{\boldsymbol{x}}_{AF} = \boldsymbol{x}_F - \hat{\boldsymbol{x}}_{AF}$, and let $\boldsymbol{z}_{AFi} = \tilde{\boldsymbol{x}}_{AFi}/\omega_{AF}^{i-1}$ and $\boldsymbol{z}_{AF} = [\boldsymbol{z}_{AF1} \quad \boldsymbol{z}_{AF2} \quad \cdots \quad \boldsymbol{z}_{AFn}]^T$, and then the error model of nth-order translational AEDO is expressed as follows:

$$\begin{aligned} \dot{\boldsymbol{z}}_{AF} = \omega_{AF}(\boldsymbol{A}_F - \boldsymbol{L}_{AF0}\boldsymbol{C}_F)\boldsymbol{z}_{AF} + \boldsymbol{B}_F \boldsymbol{F}_d^{(n)}/\omega_{AF}^{n-1} \\ - \omega_{AF}\boldsymbol{L}_{AF0}\boldsymbol{n}_F \end{aligned} \tag{17}$$

where $\boldsymbol{L}_{AF0} = [l_{AF1}\boldsymbol{I}_3 \quad l_{AF2}\boldsymbol{I}_3 \quad \cdots \quad l_{AFn}\boldsymbol{I}_3]^{\mathrm{T}}$.

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As the system composed of (11) and (12) is observable, $\boldsymbol{L}_{AF0}$ can be found to ensure that $\boldsymbol{A}_F - \boldsymbol{L}_{AF0}\boldsymbol{C}_F$ is Hurwitz. Under the condition that $\boldsymbol{A}_F - \boldsymbol{L}_{AF0}\boldsymbol{C}_F$ is Hurwitz, there exists a symmetric matrix $\boldsymbol{P}_{AF} > 0$ such that

$$(\boldsymbol{A}_F - \boldsymbol{L}_{AF0}\boldsymbol{C}_F)^{\mathrm{T}}\boldsymbol{P}_{AF} + \boldsymbol{P}_{AF}(\boldsymbol{A}_F - \boldsymbol{L}_{AF0}\boldsymbol{C}_F) = -\boldsymbol{I}_{3n} \tag{18}$$

where $\boldsymbol{P}_{AF}$ is fixed and does not vary with ω_{AF} when n is set.

Define Lyapunov function V_{AF} as follows:

$$V_{AF} = \boldsymbol{z}_{AF}^T\boldsymbol{P}_{AF}\boldsymbol{z}_{AF} \tag{19}$$

and let $\mu_{AF} = ||\boldsymbol{P}_{AF}||$ and $L_{AF}^* = \sqrt{1/3}||\boldsymbol{L}_{AF0}|| = (\sum_{i=1}^n l_{AFi}^2)^{1/2}$, and then the time derivative of V_{AF} is calculated as follows:

$$\begin{aligned}\dot{V}_{AF} \le -\omega_{AF}||\boldsymbol{z}_{AF}||[||\boldsymbol{z}_{AF}|| - 2\sqrt{3}\mu_{AF}(||\boldsymbol{F}_d^{(n)}||\omega_{AF}^{-n} \\ + L_{AF}^*||\boldsymbol{n}_F||)].\end{aligned} \tag{20}$$

When $||\boldsymbol{z}_{AF}|| > 2\sqrt{3}\mu_{AF}(||\boldsymbol{F}_d^{(n)}||\omega_{AF}^{-n} + L_{AF}^*||\boldsymbol{n}_F||)$, there is $\dot{V}_{AF} < 0$ in (20). According to $\boldsymbol{z}_{AFi} = \widetilde{\boldsymbol{x}}_{AFi}/\omega_{AF}^{i-1}$, the upper bound for the estimation error $||\widetilde{\boldsymbol{x}}_{AFi}||$ will be constrained by

$$\begin{aligned}B_{AFi} = \{\boldsymbol{r}_{AFi}|||\boldsymbol{r}_{AFi}|| \le 2\sqrt{3}\mu_{AF}(||\boldsymbol{F}_d^{(n)}||/\omega_{AF}^{n-i+1} \\ + \omega_{AF}^{i-1}L_{AF}^*||\boldsymbol{n}_F||)\}.\end{aligned} \tag{21}$$

B. Rotational AEDO

Similar to the design of translational AEDO, the states and state matrix of $\boldsymbol{\tau}_d$ can be denoted as $\boldsymbol{x}_{Ti} = \boldsymbol{\tau}_d^{(i-1)}$ and $\boldsymbol{x}_T = [\boldsymbol{x}_{T1} \quad \boldsymbol{x}_{T2} \quad \cdots \quad \boldsymbol{x}_{Tn}]^T$, respectively. Based on (2) and $\boldsymbol{\omega}_m = \boldsymbol{\omega} + \boldsymbol{n}_\omega$, the virtual measurement of $\boldsymbol{\tau}_d$ is obtained by $\boldsymbol{\tau}_{dm} = \boldsymbol{J}\dot{\boldsymbol{\omega}}_m + \boldsymbol{\omega}_m^\times\boldsymbol{J}\boldsymbol{\omega}_m - \boldsymbol{\tau}_c$. Define $\boldsymbol{C}_T = [\boldsymbol{I}_3 \quad \boldsymbol{O}_{3\times3(n-1)}]$ and the measurement noise in $\boldsymbol{\tau}_{dm}$ as $\boldsymbol{n}_T = \boldsymbol{\tau}_{dm} - \boldsymbol{\tau}_d$, and then the nth-order model of $\boldsymbol{\tau}_d$ in state space is constructed as follows:

$$\begin{cases}\dot{\boldsymbol{x}}_T = \boldsymbol{A}_T\boldsymbol{x}_T + \boldsymbol{B}_T\boldsymbol{\tau}_d^{(n)} \\ \boldsymbol{y}_T = \boldsymbol{C}_T\boldsymbol{x}_T + \boldsymbol{n}_T\end{cases} \tag{22}$$

where $\boldsymbol{A}_T = \begin{bmatrix}\boldsymbol{O}_{3(n-1)\times3} & \boldsymbol{I}_{3(n-1)} \\ \boldsymbol{O}_{3\times3} & \boldsymbol{O}_{3\times3(n-1)}\end{bmatrix}$, $\boldsymbol{B}_T = \begin{bmatrix}\boldsymbol{O}_{3(n-1)\times3} \\ \boldsymbol{I}_3\end{bmatrix}$, $i = 1, 2, \ldots, n$.

For AEDO, define the estimate of $\boldsymbol{\tau}_d$ as $\hat{\boldsymbol{x}}_{ATi} = \hat{\boldsymbol{\tau}}_{dA}^{(i-1)}$, and its estimate matrix as $\hat{\boldsymbol{x}}_{AT} = [\hat{\boldsymbol{\tau}}_{dA} \quad \hat{\boldsymbol{\tau}}_{dA}^{(1)} \quad \cdots \quad \hat{\boldsymbol{\tau}}_{dA}^{(n-1)}]^T$, and then the nth-order rotational AEDO is designed as follows:

$$\begin{cases}\dot{\hat{\boldsymbol{x}}}_{AT} = \boldsymbol{A}_T\hat{\boldsymbol{x}}_{AT} + \boldsymbol{L}_{AT}\boldsymbol{C}_T(\boldsymbol{x}_T - \hat{\boldsymbol{x}}_{AT}) + \boldsymbol{L}_{AT}\boldsymbol{n}_T \\ \hat{\boldsymbol{y}}_{AT} = \boldsymbol{C}_T\hat{\boldsymbol{x}}_{AT}\end{cases} \tag{23}$$

where $\boldsymbol{L}_{AT} = [l_{AT1}\omega_{AT}\boldsymbol{I}_3 \quad l_{AT2}\omega_{AT}^2\boldsymbol{I}_3 \quad \cdots \quad l_{ATn}\omega_{AT}^n\boldsymbol{I}_3]^{\mathrm{T}}$ is the gain matrix, $l_{ATi} > 0$, ω_{AT} is the variable bandwidth, $\hat{\boldsymbol{y}}_{AT}$ denotes the estimate of $\boldsymbol{y}_T$.

To establish the adaptive tuning rule of ω_{AT}, define the smoothed power of $\boldsymbol{\tau}_d^{(n)}$ and $\boldsymbol{n}_T$ as follows:

$$P_T = \int_{t-\Delta t}^t |\boldsymbol{\tau}_d^{(n)}(t)|^2 dt/\Delta t \tag{24}$$

$$P_{NT} = \int_{t-\Delta t}^t |\boldsymbol{n}_T(t)|^2 dt/\Delta t \tag{25}$$

and the power ratio as $\eta_T = P_T/P_{NT}$, then the adaptive tuning rule of ω_{AT} is designed as follows:

$$\omega_{AT} = \begin{cases}(K_T\eta_T)^{1/2n}, \ (K_T\eta_T)^{1/2n} > \omega_{ATm} \\ \omega_{ATm}, \text{other}\end{cases} \tag{26}$$

where $K_T = \begin{cases}1, \ \int_{t-\Delta t}^t \omega_{AT}^2 dt/\Delta t < 1 \\ \int_{t-\Delta t}^t \omega_{AT}^2 dt/\Delta t, \text{other}\end{cases}$ is used to accelerate the tuning speed, and $\omega_{ATm} = \min(\omega_{AT})$.

Similar to translational AEDO, let $\widetilde{\boldsymbol{x}}_{AT} = \boldsymbol{x}_T - \hat{\boldsymbol{x}}_{AT}$ and $\boldsymbol{z}_{ATi} = \widetilde{\boldsymbol{x}}_{ATi}/\omega_{AT}^{i-1}$, then Lyapunov function V_{AT} is designed as follows:

$$V_{AT} = \boldsymbol{z}_{AT}^T\boldsymbol{P}_{AT}\boldsymbol{z}_{AT} \tag{27}$$

where $\boldsymbol{P}_{AT}$ satisfies $(\boldsymbol{A}_F - \boldsymbol{L}_{AF0}\boldsymbol{C}_F)^{\mathrm{T}}\boldsymbol{P}_{AF} + \boldsymbol{P}_{AF}(\boldsymbol{A}_F - \boldsymbol{L}_{AF0}\boldsymbol{C}_F) = -\boldsymbol{I}_{3n}$, $\boldsymbol{P}_{AT} > 0$, and $\boldsymbol{L}_{AT0} = [l_{AT1}\boldsymbol{I}_3 \quad l_{AT2}\boldsymbol{I}_3 \quad \cdots \quad l_{ATn}\boldsymbol{I}_3]^T$.

When $||\boldsymbol{z}_{AT}|| > 2\sqrt{3}\mu_{AT}(||\boldsymbol{\tau}_d^{(n)}||\omega_{AT}^{-n} + L_{AT}^*||\boldsymbol{n}_T||)$, there is $\dot{V}_{AT} < 0$. Therefore, the upper bound for $||\widetilde{\boldsymbol{x}}_{ATi}||$ will be constrained by

$$\begin{aligned}B_{ATi} = \{\boldsymbol{r}_{ATi}|||\boldsymbol{r}_{ATi}|| \le 2\sqrt{3}\mu_{AT}(||\boldsymbol{\tau}_d^{(n)}||/\omega_{AT}^{n-i+1} \\ + \omega_{AT}^{i-1}L_{AT}^*||\boldsymbol{n}_T||)\}\end{aligned} \tag{28}$$

where $\mu_{AT} = ||\boldsymbol{P}_{AT}||$ and $L_{AT}^* = (\sum_{i=1}^n l_{ATi}^2)^{1/2}$.

C. Accuracy Analysis and Comparison

In this section, AEDO will be compared with the latest disturbance estimation methods for the landing on fast platforms in the literature, namely EDO [22] and NREDO [23], as they exhibit superior disturbance suppression performance in the landing scenario. The comparison will take the observers for the translational subsystem as an example.

1) Estimation Accuracy of AEDO: Define the estimation error of $\boldsymbol{F}_d$ as $\widetilde{\boldsymbol{F}}_{dA} = \widetilde{\boldsymbol{x}}_{A1}$, and the upper bound of $\widetilde{\boldsymbol{F}}_{dA}$ is denoted based on (21) as follows:

$$r_{AF} = 2\sqrt{3}\mu_{AF}(||\boldsymbol{F}_d^{(n)}||/\omega_{AF}^n + L_{AF}^*||\boldsymbol{n}_F||). \tag{29}$$

To intuitively reflect the effect of the adaptive tuning rule on r_{AF}, define the smoothed power of r_{AF} as follows:

$$P_{RAF} = \int_{t-\Delta t}^t r_{AF}^2 dt/\Delta t \tag{30}$$

and then P_{RAF} can be further expressed based on (29) as follows:

$$P_{RAF} \le 24\mu_{AF}^2[\omega_{AF}^{-2n}P_F + {L_{AF}^*}^2P_{NF}]. \tag{31}$$

Substitute (16) into (31), and P_{RAF} can be denoted as follows:

$$P_{RAF} \le \begin{cases}24\mu_{AF}^2(K^{-1}P_{NF} + {L_{AF}^*}^2P_{NF}), \ \omega_{AF} \ge \omega_{AFm} \\ 24\mu_{AF}^2(\omega_{AFm}^{-2n}P_F + {L_{AF}^*}^2P_{NF}), \text{other}.\end{cases} \tag{32}$$

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In (32), the upper bound of P_{RAF} depends on P_{NF} rather than both P_F and P_{NF} when $\omega_{AF} \geq \omega_{AFm}$. This implies that P_F will not affect the upper bound of P_{RAF} once ω_{AF} begins to vary according to P_F and P_{NF}. As P_{NF} decreases when the quadrotor UAV descends, the estimation accuracy of AEDO will gradually improve in the landing process according to (32).

2) Comparison With EDO: Define the estimate of $\boldsymbol{F}_d$ in EDO as $\hat{\boldsymbol{x}}_{Ei} = \hat{\boldsymbol{F}}_{dE}^{(i-1)}$ and its estimate matrix as $\hat{\boldsymbol{x}}_E = [\hat{\boldsymbol{x}}_{E1} \quad \hat{\boldsymbol{x}}_{E2} \quad \cdots \quad \hat{\boldsymbol{x}}_{En}]^T$, then the nth-order EDO is designed as follows:

$$\dot{\hat{\boldsymbol{x}}}_E = \boldsymbol{A}_E \hat{\boldsymbol{x}}_E + \boldsymbol{L}_E \boldsymbol{C}_E (\boldsymbol{x}_F - \hat{\boldsymbol{x}}_E) + \boldsymbol{L}_E \boldsymbol{n}_F \tag{33}$$

where $\boldsymbol{L}_E = [l_{E1}\omega_E \boldsymbol{I}_3 \quad l_{E2}\omega_E^2 \boldsymbol{I}_3 \quad \cdots \quad l_{En}\omega_E^n \boldsymbol{I}_3]^T$ is the gain matrix, $l_{Ei} > 0$, ω_E is the fixed bandwidth, and $\boldsymbol{C} = [\boldsymbol{I}_3 \quad \boldsymbol{O}_{3\times 3(n-1)}]$ as well as $\boldsymbol{A}_E = \begin{bmatrix} \boldsymbol{O}_{3(n-1)\times 3} & \boldsymbol{I}_{3(n-1)} \\ \boldsymbol{O}_{3\times 3} & \boldsymbol{O}_{3\times 3(n-1)} \end{bmatrix}$.

Following the deduction of r_{AF} and P_{RAF}, define the estimation error for EDO as $\widetilde{\boldsymbol{F}}_{dE} = \widetilde{\boldsymbol{x}}_{E1} = \boldsymbol{x}_F - \hat{\boldsymbol{x}}_{E1}$, and the upper bound of $\widetilde{\boldsymbol{F}}_{dE}$ as follows:

$$r_{EF} = 2\sqrt{3}\mu_E(||\boldsymbol{F}_d^{(n)}||/\omega_E^n + L_E^*||\boldsymbol{n}_F||) \tag{34}$$

and then the smoothed power of r_{EF} is denoted as follows:

$$P_{REF} \leq 24\mu_E^2(\omega_E^{-2n} P_F + {L_E^*}^2 P_{NF}) \tag{35}$$

where $L_E^* = (\sum_{i=1}^n l_{EFi}^2)^{1/2}$ and μ_E is a coefficient related to n.

According to (32) and (35), the difference between AEDO and EDO is that the upper bound of P_{RAF} will not be affected by P_F when $\omega_{AF} \geq \omega_{AFm}$. When P_F increases with the descend of the quadrotor, AEDO can maintain consistent estimation performance, whereas EDO will have decreased performance. When P_{NF} decreases in the landing process, AEDO will have higher improvement in estimation performance than EDO.

Moreover, $P_{RAF} \leq P_{REF}$ when $(K_F P_F / P_{NF})^{1/2n} \geq \omega_E$ and $\omega_{AF} \geq \omega_{AFm}$. As P_F increases, P_{NF} decreases, and ω_E is fixed when the quadrotor UAV descends, the condition can be satisfied in the landing scenario. When the condition is met, the estimation accuracy of AEDO is superior to that of EDO.

3) Comparison With NREDO: According to [23], the expanded state of $\boldsymbol{F}_d$ is denoted as $\boldsymbol{x}_{F0} = \int_0^t \boldsymbol{F}_{dm}(\tau)d\tau$, and the expanded state matrix of $\boldsymbol{F}_d$ is expressed as $\boldsymbol{x}_{NF} = [\boldsymbol{x}_{F0} \quad \boldsymbol{x}_{F1} \quad \cdots \quad \boldsymbol{x}_{Fn}]^T$. Define the estimate and estimate matrix of $\boldsymbol{F}_d$ as $\hat{\boldsymbol{x}}_{Nj} = \boldsymbol{x}_{Fj}$ and $\hat{\boldsymbol{x}}_N = [\hat{\boldsymbol{x}}_{N0} \quad \hat{\boldsymbol{x}}_{N1} \quad \cdots \quad \hat{\boldsymbol{x}}_{Nn}]^T$, then the $(n+1)$th-order NREDO is designed as follows:

$$\dot{\hat{\boldsymbol{x}}}_N = \boldsymbol{A}_N \hat{\boldsymbol{x}}_N + \boldsymbol{L}_N \boldsymbol{C}_N (\boldsymbol{x}_{NF} - \hat{\boldsymbol{x}}_N) \tag{36}$$

where $\boldsymbol{L}_N = [l_{N0}\omega_N \boldsymbol{I}_3 \quad l_{N1}\omega_N^2 \boldsymbol{I}_3 \quad \cdots \quad l_{Nn}\omega_N^{n+1} \boldsymbol{I}_3]^T$ is the gain matrix, ω_N is the bandwidth, and $\boldsymbol{A}_N = \begin{bmatrix} \boldsymbol{O}_{3n\times 3} & \boldsymbol{I}_{3n} \\ \boldsymbol{O}_{3\times 3} & \boldsymbol{O}_{3\times 3n} \end{bmatrix}$ and $\boldsymbol{C}_N = \begin{bmatrix} \boldsymbol{I}_3 \\ \boldsymbol{O}_{3n\times 3} \end{bmatrix}^T$.

Similar to r_{AF} and P_{RAF}, define the estimation error for NREDO as $\widetilde{\boldsymbol{F}}_{dN} = \widetilde{\boldsymbol{x}}_{N1}$, and the upper bound of $\widetilde{\boldsymbol{F}}_{dN}$ as follows:

$$r_{NF} = 2\sqrt{3}\mu_N(||\boldsymbol{F}_d^{(n)}||/\omega_N^n + ||\boldsymbol{n}_F||) \tag{37}$$

and the smoothed power of r_{NF} are denoted as follows:

$$P_{RNF} \leq 24\mu_N^2(\omega_N^{-2n} P_F + P_{NF}) \tag{38}$$

where $L_E^* = (\sum_{i=1}^n l_{EFi}^2)^{1/2}$, μ_N is a coefficient related to n.

According to (32) and (38), AEDO and NREDO each have their own advantage. For NREDO, the weight coefficient of P_{NF} in P_{RNF} is smaller than that in P_{RAF}. However, the reduction of P_{NF} in the landing process will weaken this improvement, and the strengthened P_F as well as the fixed ω_N will cause P_{RNF} to be larger.

In contrast, the advantage of AEDO will not be weakened in the landing process. Specifically, upper bound of P_{RAF} is only affected by P_{NF} when $\omega_{AF} \geq \omega_{AFm}$. When P_F increases and P_{NF} decreases in the landing process, the upper bound of P_{RAF} will become smaller.

Furthermore, $P_{RAF} \leq P_{RNF}$ when $\omega_{AF} \geq \omega_{AFm}$ and $[\mu_N^2 K_F P_F / P_{NF}(K_F {L_A^*}^2 \mu_A^2 + \mu_A^2 - K_F \mu_N^2)]^{1/2n} \geq \omega_N$. When the condition is met, the estimation accuracy of AEDO is superior to that of NREDO.

Remark 1: AEDO takes EDO as the basic structure and tunes the bandwidth according to the power ratio of disturbances to measurement noises. Specifically, AEDO will adopt a small bandwidth when the quadrotor UAV is far from the platform due to weak disturbances and strong measurement noises, and vice versa. Therefore, the advantage of AEDO lies in the identification and utilization the characteristics of disturbances and measurement noises in the landing process.

IV. AEDO-Based Controller

In this section, an AEDO-based block backstepping controller is designed including a position controller, an attitude controller, and its stability will be analyzed.

A. Controller Design

Define the expected value of $\boldsymbol{r}$, $\boldsymbol{q}$, and $\boldsymbol{\omega}$ as $\boldsymbol{r}_c = \boldsymbol{p}_m$, $\boldsymbol{q}_c$, and $\boldsymbol{\omega}_c$, and the error as $\boldsymbol{r}_e = \boldsymbol{r}_m - \boldsymbol{r}_c$, $\boldsymbol{q}_e = [q_{0e} \quad \boldsymbol{q}_{ve}]^T = \boldsymbol{q}_c^* \otimes \boldsymbol{q}_m$, and $\boldsymbol{\omega}_e$, then $\boldsymbol{F}_c$ and $\boldsymbol{\tau}_c$ are given by

$$\begin{cases} \boldsymbol{F}_c = -(\boldsymbol{K}_{c1}\boldsymbol{K}_{c3} + \boldsymbol{K}_{c2})\boldsymbol{r}_e - (m\boldsymbol{K}_{c1} + \boldsymbol{K}_{c3})\dot{\boldsymbol{r}}_e + m\ddot{\boldsymbol{r}}_c - \hat{\boldsymbol{F}}_{dA} \\ \boldsymbol{\tau}_c = -(1 + \boldsymbol{K}_{c4}\boldsymbol{K}_{c5})\boldsymbol{q}_{ve} - \boldsymbol{J}\boldsymbol{K}_{c4}\dot{\boldsymbol{q}}_{ve} - \boldsymbol{K}_{c5}\boldsymbol{\omega}_e - \hat{\boldsymbol{\tau}}_{dA} \\ \qquad + \boldsymbol{\omega}_m \times \boldsymbol{J}\boldsymbol{\omega}_m + \boldsymbol{J}\dot{\boldsymbol{R}}_C^B \boldsymbol{\omega}_c + \boldsymbol{J}\boldsymbol{R}_C^B \dot{\boldsymbol{\omega}}_c \end{cases} \tag{39}$$

where $\boldsymbol{K}_{c1}$, $\boldsymbol{K}_{c2}$, $\boldsymbol{K}_{c3}$, $\boldsymbol{K}_{c4}$, and $\boldsymbol{K}_{c5}$ are the gain matrices, $\hat{\boldsymbol{F}}_{dA}$ and $\hat{\boldsymbol{\tau}}_{dA}$ are obtained from AEDO, and $\boldsymbol{R}_C^B$ denotes the rotation of $\boldsymbol{q}_c$ relative to $\boldsymbol{q}$.

B. Stability Analysis

First, define $\widetilde{\boldsymbol{v}}_e = \boldsymbol{v}_e + \boldsymbol{K}_{c1}\boldsymbol{r}_e$ and $\widetilde{\boldsymbol{\omega}}_e = \boldsymbol{\omega}_e + \boldsymbol{K}_{c4}\boldsymbol{q}_{ve}$,and a candidate Lyapunov function V_C is selected as follows:

$$V_C = \boldsymbol{r}_e^{\mathrm{T}} \boldsymbol{K}_{c2} \boldsymbol{r}_e / 2 + m\widetilde{\boldsymbol{v}}_e^{\mathrm{T}} \widetilde{\boldsymbol{v}}_e / 2 + \boldsymbol{q}_{ve}^{\mathrm{T}} \boldsymbol{q}_{ve} + (1 - q_{0e})^2$$

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$$+\widetilde{\boldsymbol{\omega}}_e^{\mathrm{T}}\boldsymbol{J}\boldsymbol{J}\widetilde{\boldsymbol{\omega}}_e/2+V_{AF}+V_{AT}. \tag{40}$$

The time derivative of V_C is written as follows:

$$\dot{V}_C=\boldsymbol{r}_e^{\mathrm{T}}\boldsymbol{K}_{c2}\boldsymbol{v}_e+m\widetilde{\boldsymbol{v}}_e^{\mathrm{T}}\dot{\widetilde{\boldsymbol{v}}}_e+\boldsymbol{q}_{ve}^{\mathrm{T}}\boldsymbol{\omega}_e+\widetilde{\boldsymbol{\omega}}_e^{\mathrm{T}}\boldsymbol{J}\boldsymbol{J}\dot{\widetilde{\boldsymbol{\omega}}}_e+\dot{V}_{AF}+\dot{V}_{AT}. \tag{41}$$

Define $\boldsymbol{N}=-\boldsymbol{\omega}_m^{\times}\boldsymbol{J}\boldsymbol{\omega}_m-\boldsymbol{J}\dot{\boldsymbol{R}}_D^B\boldsymbol{\omega}_{cm}-\boldsymbol{J}\boldsymbol{R}_D^B\dot{\boldsymbol{\omega}}_{cm}$ and substitute (2) into (41), then $\dot{V}_C$ is derived as follows:

$$\begin{aligned}\dot{V}_C=&-\boldsymbol{r}_e^{\mathrm{T}}\boldsymbol{K}_{c1}\boldsymbol{K}_{c2}\boldsymbol{r}_e+\widetilde{\boldsymbol{v}}_e^{\mathrm{T}}(m\boldsymbol{K}_{c1}\dot{\boldsymbol{r}}_e+\boldsymbol{K}_{c2}\boldsymbol{r}_e\\&+\boldsymbol{F}_c+\boldsymbol{F}_d-m\ddot{\boldsymbol{r}}_c)+\dot{V}_{AF}-\boldsymbol{q}_{ve}^{\mathrm{T}}\boldsymbol{K}_{c4}\boldsymbol{q}_{ve}\\&+\widetilde{\boldsymbol{\omega}}_e^{\mathrm{T}}(\boldsymbol{J}\boldsymbol{K}_{c4}\dot{\boldsymbol{q}}_{ve}+\boldsymbol{J}\boldsymbol{q}_{ve}+\boldsymbol{\tau}_c+\boldsymbol{\tau}_d+\boldsymbol{N})+\dot{V}_{AT}.\end{aligned} \tag{42}$$

Define $H_F=\widetilde{\boldsymbol{v}}_e^{\mathrm{T}}(m\boldsymbol{K}_{c1}\dot{\boldsymbol{r}}_e+\boldsymbol{K}_{c2}\boldsymbol{r}_e+\boldsymbol{F}_c+\boldsymbol{F}_d-m\ddot{\boldsymbol{r}}_c)$ and $H_T=\widetilde{\boldsymbol{\omega}}_e^{\mathrm{T}}(\boldsymbol{J}\boldsymbol{K}_{c4}\dot{\boldsymbol{q}}_{ve}+\boldsymbol{J}\boldsymbol{q}_{ve}+\boldsymbol{\tau}_c+\boldsymbol{\tau}_d+\boldsymbol{N})$, and substitute (39) into H_F and H_T individually, then H_F and H_T can be expressed, respectively as follows:

$$\begin{aligned}H_F&=\widetilde{\boldsymbol{v}}_e^{\mathrm{T}}(-\boldsymbol{K}_{c1}\boldsymbol{K}_{c3}\boldsymbol{r}_e-\boldsymbol{K}_{c3}\dot{\boldsymbol{r}}_e+\widetilde{\boldsymbol{F}}_{dA}+\boldsymbol{n}_{cF})\\&=-\widetilde{\boldsymbol{v}}_e^{\mathrm{T}}\boldsymbol{K}_{c3}\widetilde{\boldsymbol{v}}_e+\widetilde{\boldsymbol{v}}_e^{\mathrm{T}}(\widetilde{\boldsymbol{F}}_{dA}+\boldsymbol{n}_{cF})\end{aligned} \tag{43}$$

$$\begin{aligned}H_T&=\widetilde{\boldsymbol{\omega}}_e^{\mathrm{T}}(-\boldsymbol{K}_{c4}\boldsymbol{K}_{c5}\boldsymbol{q}_{ve}-\boldsymbol{K}_{c5}\boldsymbol{\omega}_e+\widetilde{\boldsymbol{\tau}}_{dA}++\boldsymbol{n}_{cT})\\&=-\widetilde{\boldsymbol{\omega}}_e^{\mathrm{T}}\boldsymbol{K}_{c5}\widetilde{\boldsymbol{\omega}}_e+\widetilde{\boldsymbol{\omega}}_e^{\mathrm{T}}(\widetilde{\boldsymbol{\tau}}_{dA}+\boldsymbol{n}_{cT})\end{aligned} \tag{44}$$

where $\boldsymbol{n}_{cF}=m\dot{\boldsymbol{n}}_v$ and $\boldsymbol{n}_{cT}=\boldsymbol{J}\dot{\boldsymbol{n}}_\omega-\boldsymbol{n}_\omega^{\times}\boldsymbol{J}\boldsymbol{n}_\omega+\boldsymbol{\omega}_m^{\times}\boldsymbol{J}\boldsymbol{n}_\omega+\boldsymbol{n}_\omega^{\times}\boldsymbol{J}\boldsymbol{\omega}_m$.

Define λ_1, λ_2, λ_3, λ_4, and λ_5 as the minimal eigenvalues of $\boldsymbol{K}_{c1}$, $\boldsymbol{K}_{c2}$, $\boldsymbol{K}_{c3}$, $\boldsymbol{K}_{c4}$, and $\boldsymbol{K}_{c5}$, respectively, and $\dot{V}_C$ can be derived by substituting (43) and (44) into (42) as follows:

$$\begin{aligned}\dot{V}_C=&-\boldsymbol{r}_e^{\mathrm{T}}\boldsymbol{K}_{c1}\boldsymbol{K}_{c2}\boldsymbol{r}_e-\widetilde{\boldsymbol{v}}_e^{\mathrm{T}}\boldsymbol{K}_{c3}\widetilde{\boldsymbol{v}}_e+\widetilde{\boldsymbol{v}}_e^{\mathrm{T}}(\widetilde{\boldsymbol{F}}_{dA}+\boldsymbol{n}_{cF})+\dot{V}_{AF}\\&-\boldsymbol{q}_{ve}^{\mathrm{T}}\boldsymbol{K}_{c4}\boldsymbol{q}_{ve}-\widetilde{\boldsymbol{\omega}}_e^{\mathrm{T}}\boldsymbol{K}_{c5}\widetilde{\boldsymbol{\omega}}_e+\widetilde{\boldsymbol{\omega}}_e^{\mathrm{T}}(\widetilde{\boldsymbol{\tau}}_{dA}+\boldsymbol{n}_{cT})+\dot{V}_{AT}\\\le&-\lambda_1\lambda_2||\boldsymbol{r}_e||^2-\lambda_3||\boldsymbol{v}_e||^2+||\widetilde{\boldsymbol{v}}_e||(||\widetilde{\boldsymbol{F}}_{dA}||+||\boldsymbol{n}_{cF}||)\\&+\dot{V}_{AF}-\lambda_4||\boldsymbol{q}_{ve}||^2-\lambda_5||\widetilde{\boldsymbol{\omega}}_e||^2+||\widetilde{\boldsymbol{\omega}}_e||(||\widetilde{\boldsymbol{\tau}}_{dA}||\\&+||\boldsymbol{n}_{cT}||)+\dot{V}_{AT}.\end{aligned} \tag{45}$$

Define the upper bounds of $\boldsymbol{n}_{cF}$, $\boldsymbol{n}_{cT}$, $\boldsymbol{n}_F$, and $\boldsymbol{n}_T$ as δ_{ncF}, δ_{ncT}, δ_{nF}, and δ_{nT}, respectively, and substitute (20) into (45), then $\dot{V}_C$ can be denoted as follows:

$$\begin{aligned}\dot{V}_C\le&-\lambda_1\lambda_2||\boldsymbol{r}_e||^2-(\lambda_3-1/2)||\boldsymbol{v}_e||^2-||\boldsymbol{z}_{AF}||^2/2\\&+2\sqrt{3}\mu_{AF}||\boldsymbol{z}_{AT}||(\gamma_{dT}/\omega_{AF}^n+L_{AF}^*\delta_{nF})+\delta_{ncF}||\widetilde{\boldsymbol{v}}_e||\\&-\lambda_4||\boldsymbol{q}_{ve}||^2-(\lambda_5-1/2)||\widetilde{\boldsymbol{\omega}}_e||^2-||\boldsymbol{z}_{AT}||^2/2\\&+2\sqrt{3}\mu_{AT}||\boldsymbol{z}_{AT}||(\gamma_{dR}/\omega_{AT}^n+L_{AT}^*\delta_{nT})+\delta_{ncT}||\widetilde{\boldsymbol{\omega}}_e||.\end{aligned} \tag{46}$$

Let $\boldsymbol{z}_{cAF}^{\mathrm{T}}=[\boldsymbol{r}_e^{\mathrm{T}}\quad\boldsymbol{v}_e^{\mathrm{T}}\quad\boldsymbol{z}_{AF}^{\mathrm{T}}]$ and $\boldsymbol{z}_{cAT}^{\mathrm{T}}=[\boldsymbol{q}_e^{\mathrm{T}}\quad\widetilde{\boldsymbol{\omega}}_e^{\mathrm{T}}\quad\boldsymbol{z}_{AT}^{\mathrm{T}}]$ be the uniformed error vectors, and define the uniformed gains as $\eta_{cF}=\min(\lambda_1\lambda_2,\ \lambda_3-1/2,\ 1/2)$ and $\eta_{cT}=\min(\lambda_4,\lambda_5-1/2,1/2)$, and then $\dot{V}_C$ can be expressed as follows:

$$\begin{aligned}\dot{V}_C\le&\,2\sqrt{3}\mu_{AF}(\gamma_{dT}/\omega_{AF}^n+L_{AF}^*\delta_{nF})||\boldsymbol{z}_{AF}||\\&+\delta_{ncF}||\boldsymbol{z}_{AF}||-\eta_{cF}||\boldsymbol{z}_{AF}||^2\\&+2\sqrt{3}\mu_{AT}(\gamma_{dR}/\omega_{AT}^n+L_{AT}^*\delta_{nT})||\boldsymbol{z}_{AT}||\\&+\delta_{ncT}||\boldsymbol{z}_{AT}||-\eta_{cT}||\boldsymbol{z}_{AT}||^2.\end{aligned} \tag{47}$$

(a) (b)

Fig. 3. Landing procedures. (a) Quadrotor UAV tracks the landing platform horizontally before the landing. (b) Quadrotor UAV approaches the platform vertically when the landing starts.

It is clear that $\dot{V}_{AC}<0$ whenever

$$\begin{cases}||\boldsymbol{z}_{AF}||>[\delta_{ncF}+2\sqrt{3}\mu_{AF}(\gamma_{dT}/\omega_{AF}^n+L_{AF}^*\delta_{nF})]\eta_{cF}^{-1}\\||\boldsymbol{z}_{AT}||>[\delta_{ncT}+2\sqrt{3}\mu_{AT}(\gamma_{dR}/\omega_{AT}^n+L_{AT}^*\delta_{nT})]\eta_{cT}^{-1}.\end{cases} \tag{48}$$

According to (48), $\boldsymbol{r}_e$, $\boldsymbol{v}_e$, and $\widetilde{\boldsymbol{x}}_{AF}$ are bounded and converge to the bounded ball

$$\begin{aligned}B_{CAF}=\{\boldsymbol{r}|||\boldsymbol{r}||\le&[\delta_{ncF}+2\sqrt{3}\mu_{AF}(\gamma_{dT}/\omega_{AF}^n\\&+L_{AF}^*\delta_{nF})]\eta_{cF}^{-1}\}.\end{aligned} \tag{49}$$

Since $(\boldsymbol{q}_e,\widetilde{\boldsymbol{\omega}}_e)$ is a linear mapping of $(\boldsymbol{q}_e,\boldsymbol{\omega}_e)$, $(\boldsymbol{q}_e,\boldsymbol{\omega}_e)$ can converge into a bounded area. Similarly, $(\boldsymbol{q}_e,\boldsymbol{\omega}_e)$, $\widetilde{\boldsymbol{\omega}}_e$, and $\widetilde{\boldsymbol{x}}_{AT}$ are bounded and converge to the bounded ball

$$\begin{aligned}B_{CAT}=\{\boldsymbol{r}|||\boldsymbol{r}||\le&[\delta_{ncT}+2\sqrt{3}\mu_{AT}(\gamma_{dR}/\omega_{AT}^n\\&+L_{AT}^*\delta_{nT})]\eta_{cT}^{-1}\}.\end{aligned} \tag{50}$$

V. Experimental Results and Discussion

A. Experimental System and Conditions

In order to verify the effectiveness of the proposed strategy, landing experiments on a fast platform were conducted on our self-assembled quadrotor. The landing platform is shown in Fig. 3. The experimental parameters and conditions are given in Table II. For a reliable landing on the platform, the quadrotor should maintain the same heading as the platform and land on the safe landing zone with the maximum error of landing points less than 8.5 cm. Moreover, the vision system should have sufficient bandwidth to ensure continuous tracking of the landing platform.

For similar initial state and time cost in every experiment, two procedures are arranged. In the preparatory stage, the platform travels at 8.3 m/s and the quadrotor flies with the platform at a height of 1.8 m above. When the experiment starts, the quadrotor descends vertically at an expected speed of 0.4 m/s.

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TABLE II
EXPERIMENTAL PARAMETERS AND CONDITIONS

Category	Parameter	Value
Platform	Size	1.2 m × 1.2 m
	Speed	8.3 m/s
	Direction	Southeast
Quadrotor	Mass	4.2 kg
	Autopilot	PIXHAWK
	Arm length	600 mm
	Propeller size	15 inches
Environment	Maximum wind Speed	5.94 m/s

TABLE III
CONTROL SCHEME GAINS

Module	Translational Subsystem	Rotational Subsystem
Controller	$\boldsymbol{K}_{c1} = \mathrm{diag}(0.2\ 0.2\ 0.6)$	$\boldsymbol{K}_{c4} = \mathrm{diag}(1.5\ 1.5\ 0.6)$
	$\boldsymbol{K}_{c2} = \mathrm{diag}(1.8\ 1.8\ 2.6)$	$\boldsymbol{K}_{c5} = \mathrm{diag}(0.7\ 0.7\ 0.4)$
	$\boldsymbol{K}_{c3} = \mathrm{diag}(0.8\ 0.8\ 2.4)$	
2nd-EDO	$\boldsymbol{L}_E = [4\boldsymbol{I}_3\ 4\boldsymbol{I}_3]^{\mathrm{T}}$	$\boldsymbol{L}_E = [4\boldsymbol{I}_3\ 4\boldsymbol{I}_3]^{\mathrm{T}}$
3rd-NREDOs	$\boldsymbol{L}_N = [6\boldsymbol{I}_3\ 12\boldsymbol{I}_3\ 8\boldsymbol{I}_3]^{\mathrm{T}}$	$\boldsymbol{L}_N = [6\boldsymbol{I}_3\ 12\boldsymbol{I}_3\ 8\boldsymbol{I}_3]^{\mathrm{T}}$
2nd-AEDO	$P_{NF} = 0.008z_{pq}^2 + 0.028z_{pq} + 0.026$	$P_{NT} = 0.2$

Note: Bandwidths of EDOs and NREDOs are 2 rad/s.

When the landing is completed, all motors will be turned OFF. The procedures are shown in Fig. 3, where v_h and v_v are horizontal and vertical components of $\boldsymbol{v}$.

To intuitively reflect the disturbance suppression performance of AEDO, 2nd-order EDOs, 3rd-order NREDOs, and 2nd-order AEDOs are introduced, respectively, into the closed-loop system. The EDO and NREDO-based control schemes have been presented in [22] and [23]. It should be noted that although AEDOs have higher computational complexity compared with EDOs and NREDOs, PIXHAWK is used as the autopilot in the experiments, and it has sufficient computing power to ensure the real-time performance of the quadrotor system.

For brevity, "2nd-EDO," "3rd-NREDO," and "2nd-AEDO" stand for the control based on 2nd-order EDO, 3rd-order NREDO, and 2nd-order AEDO. To reduce the effects of measurement noises, $\boldsymbol{K}_{ci}$ are tuned to ensure that the bandwidth is significantly lower than the lower frequency limit of measurement noises, where $i = 1, 2, 3, 4, 5$. The gains of the controller and observers are shown in Table III. For P_{NF} and P_{NT}, they have been fitted before the experiments. ω_{AFm} and ω_{ATm} are set as 1 rad/s to reduce the effects of measurement noises and ensure basic dynamic performance. Δt is set as 0.5 s to reduce the effects of transient power fluctuations on bandwidth variation.

B. Performance of AEDO-Based Controller

The performance of the three methods is recorded. The curves of ω_{AF} and ω_{AT} are shown in Fig. 4, where z_{pq} is defined as the relative height between the platform and quadrotor. The estimate of $\boldsymbol{F}_d$ and $\boldsymbol{\tau}_d$ are shown from Figs. 5–7. The curves of $\boldsymbol{r}_e$ and $\boldsymbol{v}_e$ are shown in Fig. 8, where $\boldsymbol{r}_e = [r_{ex}\quad r_{ey}\quad r_{ez}]^T$ and $\boldsymbol{v}_e = [v_{ex}\quad v_{ey}\quad v_{ez}]^T$. For the quadrotor attitude, define the errors of roll, pitch, and yaw angles as φ_e, θ_e, ψ_e with a 3–2–1 rotation sequence, and then their cures and the curves of $\boldsymbol{\omega}_e = [\omega_{ex}\quad \omega_{ey}\quad \omega_{ez}]^T$ are depicted in Fig. 9. The statistical results are presented in Table IV.

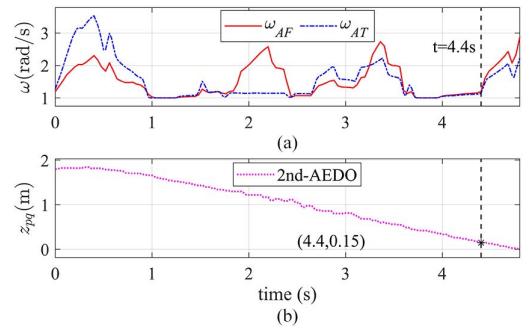

Fig. 4. Bandwidth and relative height. (a) Bandwidth of 2nd-order AEDOs. (b) Relative height between the platform and quadrotor UAV.

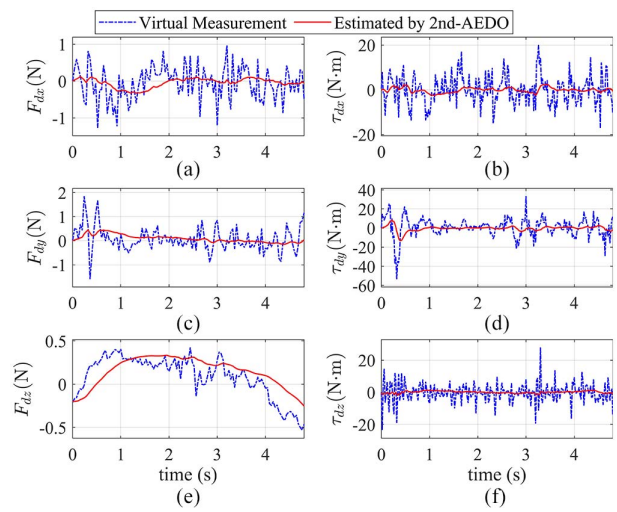

Fig. 5. Disturbance force and torque estimated by 2nd-AEDO. (a) F_{dx}. (b) τ_{dx}. (c) F_{dy}. (d) τ_{dy}. (e) F_{dz}. (f) τ_{dz}.

TABLE IV
STATISTICAL RESULTS

RMS	2nd-EDO	3rd-NREDO	2nd-AEDO
r_{ex} (m)	0.0331	0.0290	0.0252
r_{ey} (m)	0.0263	0.0294	0.0235
r_{ez} (m)	0.0777	0.0502	0.0381
v_{ex} (m/s)	0.1611	0.1253	0.1068
v_{ey} (m/s)	0.0922	0.1185	0.0558
v_{ez} (m/s)	0.1393	0.0858	0.0805
φ_e (deg)	0.5936	0.7814	0.4217
θ_e (deg)	0.9498	1.1100	0.7945
ψ_e (deg)	0.1097	0.1230	0.0348
ω_{ex} (deg/s)	2.6192	2.8090	2.6179
ω_{ey} (deg/s)	7.0915	9.0909	3.7346
ω_{ez} (deg/s)	1.1710	1.4671	1.0819

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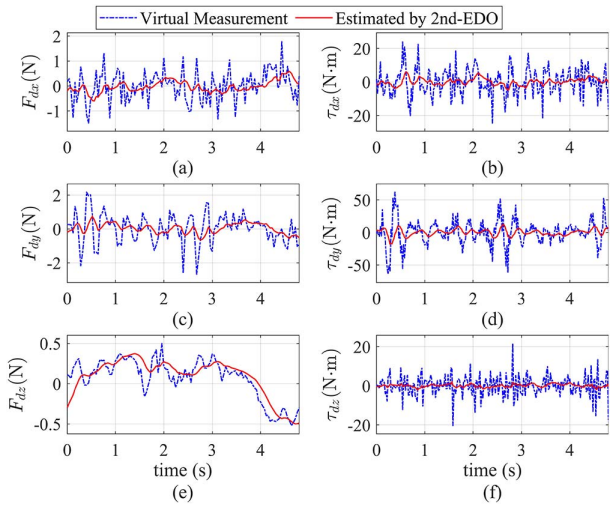

Fig. 6. Disturbance force and torque estimated by 2nd-EDO. (a) F_{dx}. (b) τ_{dx}. (c) F_{dy}. (d) τ_{dy}. (e) F_{dz}. (f) τ_{dz}.

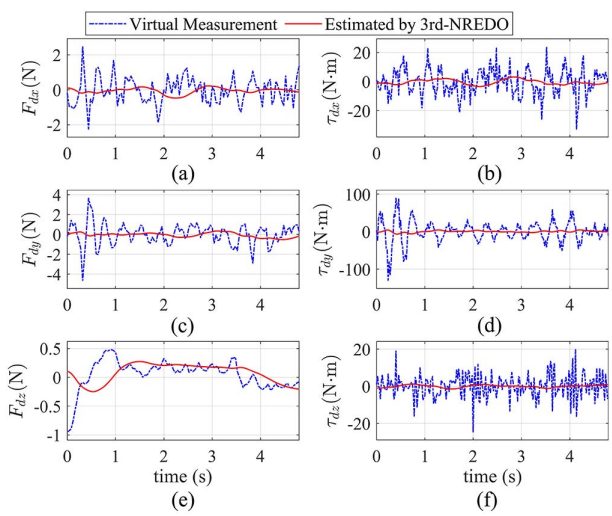

Fig. 7. Disturbance force and torque estimated by 3rd-NREDO. (a) F_{dx}. (b) τ_{dx}. (c) F_{dy}. (d) τ_{dy}. (e) F_{dz}. (f) τ_{dz}.

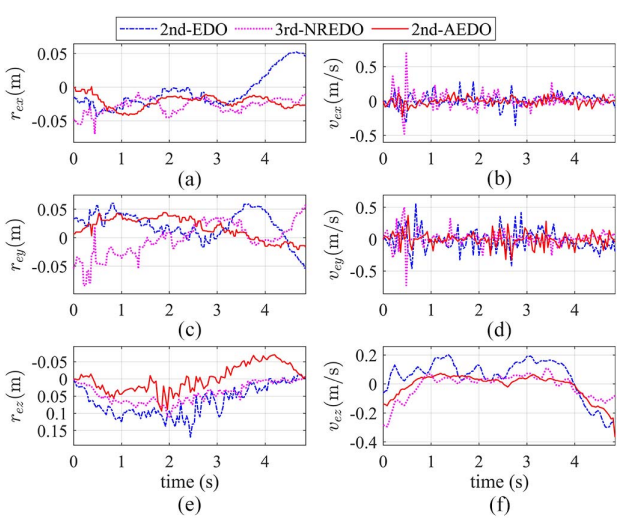

Fig. 8. Errors of position and velocity. (a) r_{ex}. (b) v_{ex}. (c) r_{ey}. (d) v_{ey}. (e) r_{ez}. (f) v_{ez}.

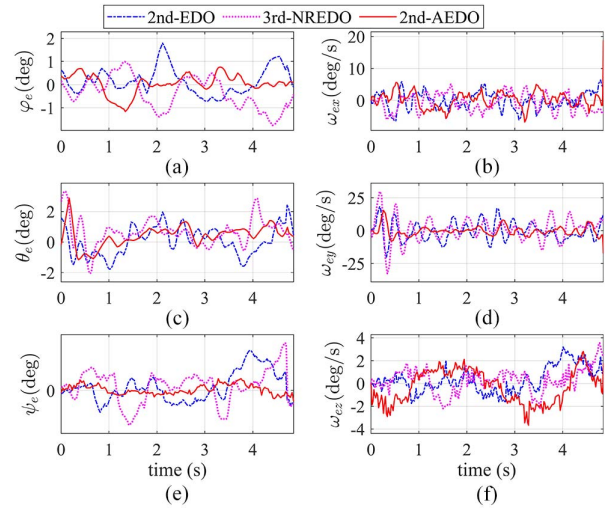

Fig. 9. Errors of attitude and angular velocity. (a) φ_e. (b) ω_{ex}. (c) θ_e. (d) ω_{ey}. (e) ψ_e. (f) ω_{ez}.

In Fig. 4, the quadrotor transitions from level to descending flight within 0–1 s. In this phase, aerodynamic disturbances increase and cause variations of ω_{AF} and ω_{AT}. Between 1.5 and 3.5 s, z_{pq} decreases from 1.4 to 0.5 m. In this phase, uneven ground effects have not occurred, and the changes of ω_{AF} and ω_{AT} are caused by wind gusts. At 4.4 s, $z_{pq} = 0.15$ m and uneven ground effects manifest. After 4.4 s, ω_{AF} and ω_{AT} increase under the combined effects of uneven ground effects and wind gusts. As P_{NF} decreases with z_{pq}, the peak of ω_{AF} continually increases. As P_{NT}does not vary with z_{pq}, the changes in the peak of ω_{AT} are small after 1 s.

In Figs. 6 and 8, the curves of 2nd-AEDO are smooth and can respond more quickly to the changes of disturbances. In Figs. 9 and 10, the curves of 2nd-AEDO exhibit the best suppression of disturbances and its curves have the lowest fluctuation. In Table III, the RMSEs of 2nd-AEDO are the smallest. This demonstrates that 2nd-AEDO has the best disturbance suppression performance among the three methods.

C. Precision of Landing Points

To ensure that the differences in disturbance estimation and suppression performance among the three methods are not random, their landing point precision will be compared in this section. The results of ten repeated experiments for the three methods are shown in Fig. 10 and Table V. In Fig. 10, the solid line circle indicates the boundary of the safe landing zone. The other line styles represent the circular ranges of landing points through different methods, and the points with various styles indicate the landing points from various methods. In Table V, (r_{ex}^P, r_{ey}^P) means the landing point is expressed in E_P. p_{cir} is the center of the circle, and r_{cir} denotes the radius.

In Fig. 10 and Table V, some landing points of 2nd-EDO and 3rd-NREDO are outside the safe landing zone while all the landing points of 2nd-AEDO are within the zone. The p_{cir} of 2nd-AEDO is the closest to zero among the three methods.

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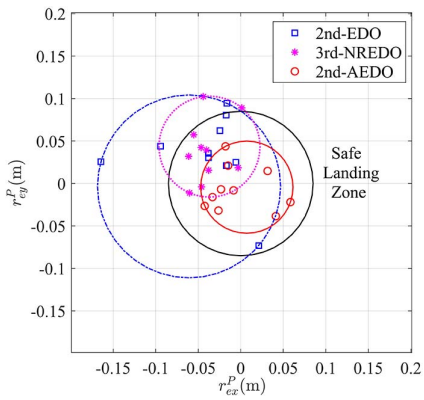

Fig. 10. Landing point of 30 times experiments.

TABLE V
RESULTS OF LANDING POINTS (M)

Variables	2nd-EDO	3rd-NREDO	2nd-AEDO
$(r_{ex}^P,\ r_{ey}^P)$	(0.0304, −0.0379)	(0.0186, −0.0030)	(−0.0081, −0.0085)
	(0.0355, −0.0381)	(0.0156, −0.0378)	(−0.0068, −0.0232)
	(0.0211, −0.0170)	(−0.0039, −0.0462)	(0.0211, −0.0149)
	(0.0250, −0.0057)	(0.0395, −0.0403)	(0.0148, 0.0316)
	(0.0622, −0.0245)	(−0.0112, −0.0602)	(−0.0160, −0.0332)
	(0.0439, −0.0943)	(0.0423, −0.0466)	(−0.0318, −0.0262)
	(0.0805, −0.0172)	(0.0320, −0.0612)	(0.0438, −0.0179)
	(0.0946, −0.0162)	(0.0574, −0.0552)	(−0.0264, −0.0424)
	(−0.0732, 0.0211)	**(0.0891, 0.0012)**	(−0.0384, 0.0412)
	(0.0257, −0.1644)	**(0.1024, −0.0440)**	(−0.0219, 0.0584)
p_{cir}	(−0.0608, −0.0033)	(−0.0368, 0.0434)	(0.0071, −0.0043)
r_{cir}	0.1076	0.0594	0.0542

Note: Bold numbers represent landing points out of the safe landing zone.

As for r_{cir}, 2nd-AEDO has the smallest r_{cir}, and the r_{cir} of 2nd-EDO and 3rd-NREDO are almost the same.

In summary, the aforementioned experimental results further validate the superior performance of AEDO in disturbance estimation and suppression and are consistent with the theoretical analysis in Section IV. Moreover, the increased computational complexity by AEDO does not affect the real-time performance of the autopilot in the experiments.

VI. CONCLUSION

This article proposes AEDO to estimate complex aerodynamic disturbances when quadrotor UAVs land on fast platforms. Two AEDOs are designed to estimate force and torque disturbances for translational and rotational channels of a quadrotor UAV. In AEDO, the bandwidth is designed to be tuned adaptively according to the power ratio between disturbances and measurement noises. The comparison with other EDOs shows that AEDO exhibits higher estimation accuracy and robustness to measurement noises. Repeated experiments on a 30 km/h platform have validated the effectiveness of the proposed strategy. In the experiments, the quadrotor UAV consistently lands on the safe landing zone with an error of less than 6.3 cm. As the increased computational complexity of the AEDO may affect its application in low-cost quadrotor UAVs, future work will focus on the reduction of the algorithm complexity.

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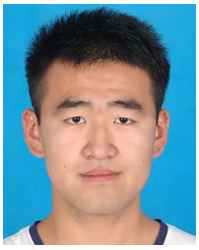

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